Vision-Based Road Rally Crowd Analysis and Agent-Based Simulation

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Semester Project Research Paper

***Abstract*—**Road rallies, protests, and other public gatherings can change how pedestrians move through urban streets, often causing congestion and limiting road access. Manual observation and ordinary video monitoring make these changes difficult to measure consistently. This paper proposes a vision-based framework for analyzing pedestrian movement during road rallies and for exploring alternative crowd conditions through simulation. A pretrained YOLO model is planned for pedestrian detection and ByteTrack for maintaining pedestrian identities across video frames, so that pedestrian counts, positions, movement directions, approximate speeds, and crowd density patterns can be extracted. This information will then be used to initialize and calibrate an agent-based simulation in which individual pedestrian agents follow simple movement rules and avoid one another and obstacles. Three scenarios are planned: normal pedestrian movement, increased crowd density, and a blocked road section. They will be compared in terms of pedestrian movement, average simulated speed, congestion, and simulated travel time. This is a proposed undergraduate semester project, and the paper presents its design and research background rather than experimental results. The expected contribution is a simple, reproducible workflow that links video-based observation with scenario-based crowd simulation.

Keywords*—Crowd analysis; pedestrian detection; multi-object tracking; agent-based modeling; YOLO; ByteTrack*

1. Introduction

Road rallies, protests, and other public gatherings are a regular part of urban life, and they can change the way people use streets within a short time. When a large group occupies a road, nearby pedestrians may slow down, change direction, or look for alternative routes, and parts of the road may become hard to access. Understanding crowd movement in such situations is relevant to public safety, event planning, and traffic management, and it has been studied across several disciplines (Zhan et al., 2008).

In practice, however, analyzing crowd movement is not simple. Manual observation depends on the attention of the observer and is difficult to repeat consistently, particularly when many people move at the same time. Ordinary video monitoring records what happens, but it does not by itself provide measurements such as the number of pedestrians, their speeds, or how density changes along a road. Automated methods that turn video into structured movement information are therefore valuable.

Computer Vision offers such methods. Pedestrian detection locates people in each video frame, and a large evaluation of detectors has shown that detection becomes harder at low resolution and when people are partly hidden (Dollár et al., 2012). YOLO is a single-network detector that predicts bounding boxes and class probabilities directly from an image, which makes it fast enough for video use (Redmon et al., 2016). Detection alone treats every frame separately, so a tracker is needed to link detections over time. ByteTrack does this by also matching low-confidence detections, which often belong to partly hidden people, to existing tracks (Zhang et al., 2022).

Observing movement is only one part of the problem, because a video shows only the situation that actually occurred. Agent-based modeling (ABM) describes a system as a set of autonomous

agents whose interactions produce collective behavior (Bonabeau, 2002). In pedestrian simulation, each agent represents a person who moves toward a goal, avoids collisions with other agents, and steers around obstacles. The social force model is a well-known example, in which pedestrian motion is described as if people were influenced by attractive and repulsive "social forces" (Helbing & Molnár, 1995). Such models allow conditions to be changed and the resulting behavior to be observed.

This paper presents a proposed system that combines these two approaches. Video of a road rally will be processed with YOLO and ByteTrack to estimate pedestrian positions, directions, and approximate speeds, and these values will be used to initialize or calibrate a simplified agent-based simulation. Three scenarios are planned: normal movement, increased crowd density, and a blocked road section. Figure 1 presents the proposed workflow; it describes a planned architecture and not an implemented or validated system. The overall aim is to compare how the three conditions may affect pedestrian movement, congestion, and simulated travel time. The project is important as a practical, low-cost example of how video analysis and simulation can support the study of road-crowd situations at the undergraduate level.

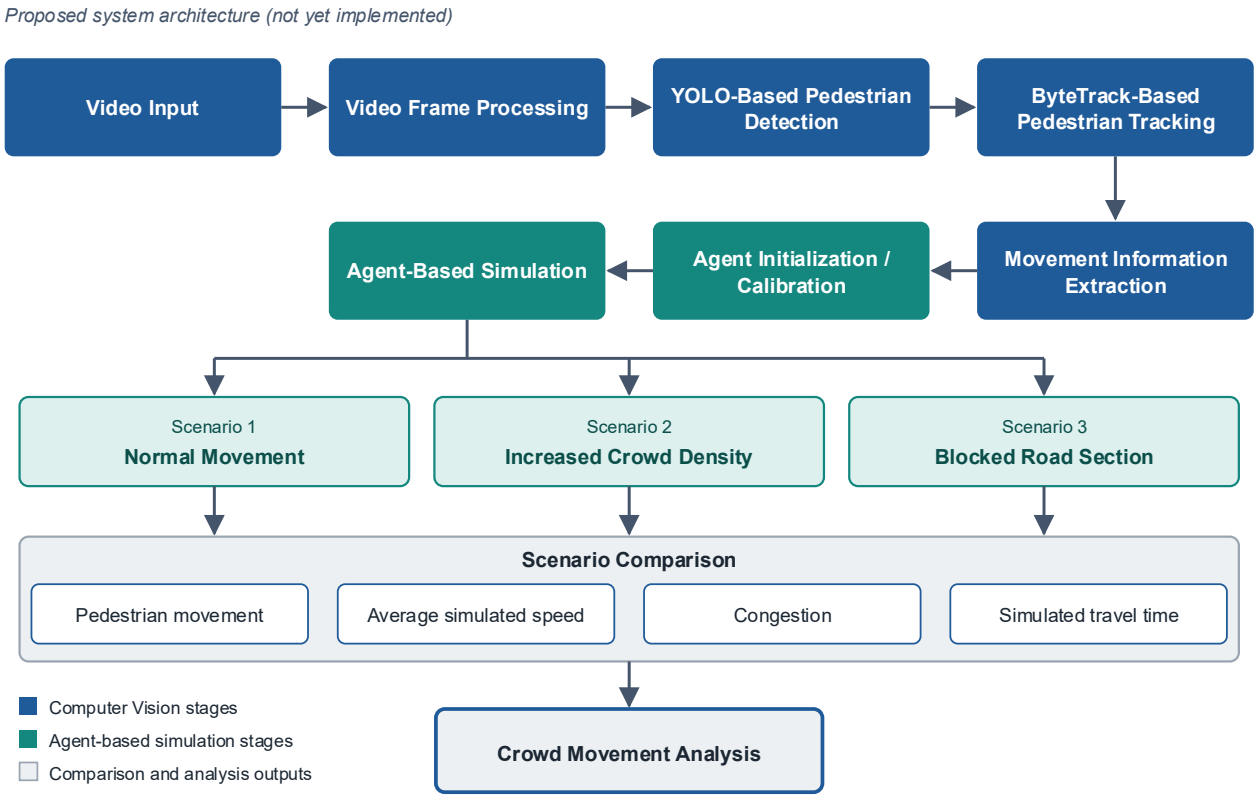


Figure 1. Proposed workflow of the Vision-Based Road Rally Crowd Analysis and Agent-Based Simulation System.

Note. The figure shows a proposed system architecture. The system has not yet been implemented or tested.

2. Literature Review

2.1 Computer Vision for Pedestrian Detection

Pedestrian detection is a long-standing problem in Computer Vision and a basis for many surveillance and traffic applications. Dollár et al. (2012) evaluated sixteen pretrained detectors across six data sets and found that performance remained weak at low resolutions and for partially occluded pedestrians. YOLO treats detection as a single regression problem from image pixels to bounding boxes and class probabilities, and its authors reported real-time operation, at the cost of more localization errors than some other systems (Redmon et al., 2016). This balance makes YOLO attractive for processing rally videos with a pretrained model.

2.2 Multi-Object Tracking

Detection treats each frame independently, so it cannot show whether a person in one frame is the same person in the next. Multi-object tracking addresses this by estimating the bounding boxes and identities of objects across a video (Zhang et al., 2022), which is necessary for measuring speed and direction. SORT combines a Kalman filter with the Hungarian algorithm for fast online association and showed that detection quality strongly affects tracking performance (Bewley et al., 2016). ByteTrack extends this idea by also associating low-score detection boxes with existing tracks instead of discarding them, which helps recover occluded people and reduces fragmented trajectories.

2.3 Crowd Movement and Density Analysis

Crowd analysis from video was surveyed by Zhan et al. (2008), who discuss tasks such as measuring crowd density and motion for crowd management and surveillance. Density can be estimated directly from images; Zhang et al. (2016) proposed a multi-column convolutional network that maps an image to a density map and copes with variation in head size caused by perspective. Tracked trajectories can instead give pedestrian flow, direction, and speed. Crowded scenes are difficult because pedestrians overlap, occlude one another, and appear at very different sizes, so counts and trajectories remain estimates (Dollár et al., 2012; Zhan et al., 2008).

2.4 Agent-Based Modeling for Crowd Simulation

Agent-based modeling describes a system from the point of view of its individual units, and even simple agents can produce complex collective patterns (Bonabeau, 2002). For pedestrians, the social force model of Helbing and Molnár (1995) represents each person as moving toward a desired direction while being repelled by other pedestrians and by obstacles. Helbing et al. (2000) used a model of this kind to study panic and jamming caused by uncoordinated motion in crowds, showing that crowd behavior at exits can differ sharply from orderly movement. These studies show how local rules for interaction and obstacle avoidance can reproduce congestion that emerges as density rises. Route changes can be represented by giving agents alternative goals when a path is blocked. A simplified road-rally model would need its own clearly stated assumptions.

2.5 Integration of Computer Vision and Agent-Based Modeling

Combining observed data with simulation allows a model to rest on real behavior instead of assumed values. Johansson et al. (2007) used video recordings of pedestrian motion and tracking software to find suitable parameter values for the social force model through evolutionary optimization, and then applied the calibrated model to simulations of evacuations, pilgrimage, and urban environments. Computer Vision can supply information about how pedestrians actually move, such as positions, directions, and speeds, while ABM can be used to examine situations that were not recorded, including higher crowd density or a blocked road section.

2.6 Research Gap

Video-based pedestrian analysis and agent-based crowd simulation are both well-established research areas, and earlier work has already linked video data with pedestrian models (Johansson et al., 2007). This project therefore does not claim to be the first such combination. Its contribution is narrower. It proposes a simplified pipeline that uses a pretrained YOLO detector and ByteTrack to obtain movement information from road-rally video for a basic agent-based simulation that compares three road-crowd scenarios: normal movement, increased density, and a blocked road section. The expected value is a clear, reproducible undergraduate-level workflow rather than new state-of-the-art performance.

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